§8 Organizational ACS: structural application to social organizations (single part)

Source: `paper/section8_organizational_acs.tex`, lines 1–445 (`sec:org_acs`: opening, `subsec:org_terms`, `subsec:org_structure`, `subsec:org_formal`, `subsec:org_viability`, `subsec:org_identity`, `subsec:org_scope`), at commit e6e12cd. Status: draft. §8 is short and is summarized in one file. Part 5 of §7 is `07_collapse_5.md`. §8 defines no assumption. The Japanese manuscript is authoritative; this is a summary, not a translation, and adds no definition, assumption or result.

Opening

`Reference: sec:org_acs`

The section takes as its object a social organization consisting of several participating agents, and describes the case in which the formal structure of an ACS holds for the organization itself. That an ACS participates in an organization is distinguished from the organization itself being an ACS. To treat the latter, the section specifies an organization-level structure and clarifies the conditions under which the definitions of earlier sections can be applied. The section deals with the abstract structure of an organization that is already identified. It does not treat the origin of the organization, the process of organizing, or any necessity of its formation. An organization's own purpose, the shared intention of its participants, and a particular form of governance are not general requirements for an organizational ACS. The distinctions of the section serve both the description of relation objects of an individual ACS and the description of the organization's own feasibility. The main new symbols are listed in `app:symbols_s8`.

Group, organization and society; organizational ACS

`Reference: subsec:org_terms`

Definition (`def:org_terms`): descriptive distinction of group, organization and society.

- A *group* is a substrate that treats several individuals as one object set under an identification rule. The definition of a group does not include, as a requirement, functional integration of the group itself or the holding of a self–environment boundary or ACS structure.
- An *organization* is a system in which several participating agents, together with contributions of non-agent elements as needed, are connected, divided in role and coordinated so as to realize organization-level functions under a distinguished organizational boundary. "Organization" here means a social organization of participating agents, not a biological tissue or a material structure. The manner of decomposing, dividing and coordinating functions depends on the implementation. The description of an organization does not presuppose a well-formed ACS structure of the organization itself.
- A *society* is the place to which individual ACSs and other agents, groups and organizations belong and in which they act. It may include these, other elements, and the relations among them.

Definition (`def:org_class_terms`): Group of ACSs, ACS organization, organizational ACS.

- A group of two or more individual ACSs is called a *group of ACSs*. Its definition does not include, as a requirement, that the organizational structure of `def:org_terms` holds.
- An organization in which individual ACSs participate is called an *ACS organization*.
- An organization for which the applicability condition of `def:org_acs` holds for the organization itself is called an *organizational ACS*.
- The definition of an ACS organization does not include, as a requirement, that the applicability condition of `def:org_acs` holds, and the definition of an organizational ACS does not include participation of individual ACSs.
- For an organization $\mathcal{O}$ treated as an organizational ACS at time t, one whose participating-ACS number (`eq:org_participants`) satisfies $n_{\mathrm{org},t}(\mathcal{O}) \geq 1$ is called an *organizational ACS*

with individual-ACS participation, and one with $n_{\mathrm{org},t}(\mathcal{O}) = 0$ an *organizational ACS without individual-ACS participation*. This division is by the record of participation and adds no condition for being an organizational ACS. The organization itself is not counted among its own participating ACSs.

Unit of description (`rem:org_acs_unit`). The organizational ACS of this paper takes an organization of several participating agents (`def:org_terms`) as the smallest unit of description at the organization level. Participating agents need not be individual ACSs. An organization that has an organization or an organizational ACS as a constituent is distinguished from this and is not treated in Phase 1.

Organizations and the classification of relation objects (`rem:org_terms_nonimplication`). The map $\mathrm{type}_i : \mathcal{Y}_{-i} \to \{\mathrm{ACS}, \mathrm{agent}, \mathrm{obj}\}$ of §6 is a classification of objects for relation description relative to an individual ACS. Because the set of external objects $\mathcal{Y}_{-i}$ does not exclude organizations and organizational ACSs, these may be classified as ACS or agent as acting agents. The value of type_i does not express whether the time-indexed applicability condition of `def:org_acs` holds. The details of describing an organizational ACS as a relation object, in particular the application to organizational ACSs of the bidirectional composition that presupposes treatment as an ACS individual (`ass:bidirect_scope`), are not treated in Phase 1 and are left to Phase 1.α or Phase 2 (`subsec:phase1a`).

Organization-level description and participating ACSs

`Reference: subsec:org_structure`

The state and function at the organization level are specified relative to the boundary of the organization concerned. From the mere enumeration of states of individual ACSs, the existence of relation records between individuals, or the presence of job titles in an organization, this paper does not construct the organization's own state, cognitive mechanism or update structure.

Definition (`def:org_participation`): time-indexed record of participating ACSs. Let $\mathbb{I}$ be the set of identifiers of individual ACSs under consideration and $\mathfrak{O}$ the set of organizations under consideration; finiteness of $\mathbb{I}$ is not assumed. For $\mathcal{O} \in \mathfrak{O}$ and time t, let $\mathbb{I}_{\mathrm{org},t}(\mathcal{O}) \subseteq \mathbb{I}$ be the set of identifiers of the distinct individuals that participate in the organization as ACSs at that time. It is called the participating-ACS set at time t, and the participating-ACS number is its cardinality

$$n_{\mathrm{org},t}(\mathcal{O}) := |\mathbb{I}_{\mathrm{org},t}(\mathcal{O})| \qquad (\texttt{eq:org_participants}).$$

$\mathbb{I}_{\mathrm{org},t}(\mathcal{O}) = \emptyset$ is allowed. This is a record made where the distinction of individuals and their participation at that time are specified; it is not assumed that ordinary ACS states of all candidate individuals exist at all times. In a model with a fixed finite candidate set, one may additionally put $P \subseteq \mathbb{I}$, $|P| < \infty$, $\mathbb{I}_{\mathrm{org},t}(\mathcal{O}) \subseteq P$ (`eq:org_candidate_pool`). No lower bound is placed on $|P|$, and the definition of the participating number does not need the existence of a fixed candidate set P.

Members and bearers of functions (`rem:org_participants_functions`). Functions of an organization may be borne by participating ACSs and also by non-ACS agents, devices, stored information and other elements. The participating-ACS number is different from the assignment of roles and effective fulfilment of functions. A common purpose or intention of all participating ACSs, fixed posts and a particular hierarchy are not required, and expressing fulfilment of a function by a head-count threshold of participating ACSs is not a general requirement either. Supply factors outside the organization's boundary are distinguished from function bearers inside it; if the former are required for the organization's persistence, there is room to describe them as required environmental factors of §2. The distinction between participating agents and non-agent elements, and whether each participating agent is an individual ACS, are specified in the target model; this paper gives no procedure that judges the distinction in general.

Relation records and organizational function (`rem:org_relation_scope`). The relation descriptors of §6 may describe relations between a participating ACS and other objects. This paper does not state, from a relation record being non-empty alone, the organization's own control mechanism, causal integration or ACS status as a general consequence. A change in a relation record is not interpreted as a unique causal contribution of another party.

Organization-level ACS structure and the applicability convention

Reference: subsec:org_formal

Definition (`def:org_structural_form`): organization-level ACS structure. Taking the organization $\mathcal{O}$ itself as the object of description, the state space, action space, required environmental factors, transition, persistence evaluation, mechanism and update are given as the 7-tuple

$$\Sigma_{\mathrm{org}} := (X_{\mathrm{org}}, U_{\mathrm{org}}, E_{\mathrm{org}}, f_{\mathrm{org}}, \Pi_{\mathrm{org}}, M_{\mathrm{org}}, \mathrm{Update}_{\mathrm{org}}) \quad \texttt{(eq:org_7tuple)}.$$

When this 7-tuple satisfies the ACS definition, the constitutive requirements and well-formedness of §2 and describes the organization's own self–environment boundary and realization of functions, it is called an *organization-level ACS structure*. In particular, $X_{\mathrm{org}} \subseteq \mathbb{R}^d$ is a non-empty compact organizational state space, and the external environment state space $E_{\mathrm{ext,org}}$ and the set of required environmental factors E_{org} are distinguished, with types

$$E_{\mathrm{org}} \subseteq \{0,1\}^{X_{\mathrm{org}} \times E_{\mathrm{ext,org}}},\ E_{\mathrm{org}} \neq \emptyset \quad \texttt{(eq:org_environment_type)},$$
$$f_{\mathrm{org}} : X_{\mathrm{org}} \times U_{\mathrm{org}} \times E_{\mathrm{ext,org}} \to X_{\mathrm{org}} \quad \texttt{(eq:org_transition_type)}.$$

$\Pi_{\mathrm{org}} : X_{\mathrm{org}} \to \mathbb{R}$ and $U_{\mathrm{org}} \neq \emptyset$; the other types, the constitutive requirements CR-1 and CR-2, and the conditions of well-formed realization follow §2. These conditions are not treated as satisfied merely by assigning symbols.

Single-valued transition (`rem:org_state_closure`). The organizational state $x_{\mathrm{org}} \in X_{\mathrm{org}}$ is not in general determined as a sum, average or product of the states of individual ACSs. In a model that uses an aggregation map, one must check that descriptions corresponding to the same organizational state, action and external environment state do not give different organizational next states; otherwise the single-valued transition of `eq:org_transition_type` is not determined. A model that gives the organizational state directly does not need an aggregation map.

Definition (`def:org_acs`): descriptive applicability convention for the organizational ACS. An organization $\mathcal{O}$ (`def:org_terms`) is treated as an *organizational ACS* at time t only if the well-formed ACS structure of `def:org_structural_form` is realized at time t for that organization itself and the objects of boundary, function and update are identified as that organization. The participating-ACS number $n_{\mathrm{org},t}(\mathcal{O})$ is not part of this condition. The specification of organization-level structure is not satisfied merely by renaming the activity of one participating agent (including an individual ACS) as the organization. A model-independent formal predicate that judges non-reducibility to individual ACSs is not introduced in this convention. The convention does not claim the general existence of organizational ACSs or that any particular real organization qualifies.

Holding versus observation (`rem:org_acs_observability`). Observation of a boundary, continuity, interaction with the environment and changes of internal function can be grounds for considering an organization-level ACS structure. This paper does not state all the conditions of `def:org_acs` as a general consequence from these observed facts alone. When internal information cannot be observed, being unable to confirm ACS status is distinguished from ACS status being absent.

Individual level versus organization level (`rem:org_individual_nonimplication`). The truth of feasibility and of the time-indexed satisfaction predicates of the three principles for participating individuals, and the same for the organization's own ACS status, feasibility and predicates, are judgments about different state spaces and evaluation objects. A general rule that takes a simple aggregation of individual quantities as an organizational quantity is not

given, and a general implication between the individual and organization levels is not claimed without an additional connecting condition.

Organization-level feasibility and the three principles

Reference: subsec:org_viability

Definition (def:org_viable_action): organization-level feasible set and feasible action set. When a well-formed organization-level ACS structure of def:org_structural_form is given,

$$K_{\mathrm{org}} := \{x \in X_{\mathrm{org}} \mid \Pi_{\mathrm{org}}(x) \geq 0\} \qquad \text{(eq:org_viable_set)}.$$

When an ordinary organization-level ACS state $x_{\mathrm{org},t} \in X_{\mathrm{org}}$ is realized at time t and $e_{\mathrm{org},t} \in E_{\mathrm{ext,org}}$ is given, the feasible action set is

$$A_{\mathrm{org},t} := \{u \in U_{\mathrm{org}} \mid f_{\mathrm{org}}(x_{\mathrm{org},t}, u, e_{\mathrm{org},t}) \in K_{\mathrm{org}}\}$$
$$\text{(eq:org_viable_action)}.$$

This set expresses the possibility of a transition to a feasible state one time step ahead and does not establish indefinite persistence.

Proposition (prop:org_blockage_characterization): characterization of organization-level blockage. If the quantities of def:org_viable_action are well-formedly defined at time t, then

$$A_{\mathrm{org},t} = \emptyset \iff \forall u \in U_{\mathrm{org}},\ \Pi_{\mathrm{org}}(f_{\mathrm{org}}(x_{\mathrm{org},t}, u, e_{\mathrm{org},t})) < 0$$
$$\text{(eq:org_blockage_characterization)}.$$

Proof idea (as in the manuscript). By eq:org_viable_set and eq:org_viable_action, $u \in A_{\mathrm{org},t}$ is equivalent to a non-negative value of Π_{org} after the transition. The absence of a feasible action is the failure of this non-negativity for every u, and since Π_{org} is real-valued, its negation is the strict negativity of eq:org_blockage_characterization.

Feasibility versus qualification (rem:org_viability_vs_qualification). Membership in K_{org} and the time-indexed qualification as an organizational ACS of def:org_acs are different predicates. The feasible set is defined from Π_{org}, and the qualification is determined by time-indexed well-formed realization. Claiming an implication between them needs a connecting condition between organizational state and transition and well-formed realization. This paper does not state qualification as an organizational ACS after the transition from $A_{\mathrm{org},t} \neq \emptyset$ alone.

Application of earlier definitions to the organization level (rem:org_principles_conditional). By def:org_structural_form the organization-level structure Σ_{org} satisfies the ACS definition, constitutive requirements and well-formedness of §2, and is therefore one realization of the abstract structure type (the convention on levels of description in subsec:acs_7tuple). Hence blockage, functional termination and collapse of §2; the time-indexed satisfaction predicates of the three principles SM, EU and FR and T-STAB of §3; and the state bands, external recoverability, trajectory records and path sets of §7 apply to Σ_{org} as they stand, with the individual index replaced by the organization $\mathcal{O}$. For example, the SM condition at the present state can be written $\Pi_{\mathrm{org}}(x_{\mathrm{org},t}) \geq 0$ (eq:org_SM). The conditions (SC-1), (SC-2), (FT-1), (FT-2) and the judgment rules of external recovery are fixed per target system, as in the individual case; for an organization, the organizational relations that establish its identity and the criterion for the identity of the same organization belong to those judgment rules. The "ACS individual" of §2 refers to a realization of the abstract structure type and is distinguished from the "individual ACS" that denotes a participant in this section; an organizational ACS is one of the ACS individuals in the former sense. On the other hand, no rule is given that transfers or aggregates quantities of participating individual ACSs (satisfaction of one-step critical required factors, truth of local FR, blockage, collapse and so on) into organization-level quantities. That the definitions can be applied to an organization differs from the principles actually being satisfied at each time and on each path. What follows from an organization-level time-indexed satisfaction predicate being false is limited to the consequences shown by the propositions of §3 (for example blockage from falsity of EU, prop:eu_blockage); loss of organizational-ACS qualification or collapse is not concluded from it alone.

Qualification, temporal identity and the scope of failure states

Reference: subsec:org_identity

Time-indexed qualification and continuity of the organization (rem:org_qualification_loss). This paper does not conclude loss of qualification as an organizational ACS under def:org_acs, loss of the organization's structural self, or collapse from a change in the participating-ACS number alone (for example a decrease of $n_{\mathrm{org},t}(\vec{\sigma})$ or its reaching 0). The case in which the organization's own ACS structure is actually lost through the withdrawal of participants is not excluded; that judgment is made on the well-formed realization at the organization level. Loss of qualification as an organizational ACS and collapse of the organization are separate judgments (rem:org_failure_scope). If the qualification fails and later holds again, it cannot be said by that alone that the same organizational ACS has returned. Where the return as the same organization is treated, inheritance from the former organization is checked separately against the continuity conditions of boundary, function and update that the target model admits. No uniform deadline for return is set. Agreement of names, or the recognition of internal participants or external observers alone, is not taken as a sufficient condition for structural identity.

Scope of blockage and collapse (rem:org_failure_scope). At times at which the organization's own ordinary ACS state and eq:org_viable_action are defined, prop:org_blockage_characterization lets one judge emptiness of the organization-level feasible action set. This is the blockage of §2 (def:blockage_sec2) applied to Σ_{org}. Blockage, departure from the feasible set, loss of well-formed realization and loss of organizational-ACS qualification are not identified. Organization-level collapse is judged by the conditions (SC-1) and (SC-2) of def:collapse of §2, under the judgment rules of rem:org_principles_conditional and the set of admissible external actions fixed for the organization (§7). Organizational-ACS qualification and the participating-ACS number are not part of the definition of collapse, so collapse of the organization is not concluded from the withdrawal or collapse of one participating ACS or from a decrease of the participating-ACS number alone. Moreover, a case in which the successor realizations that can be associated as continuation of the same organization become several, such as a split of an organization, lies outside the object of ass:no_branching of §2.

Limits of application

Reference: subsec:org_scope

The section describes the conditions under which the structural definition of an ACS can be applied to an existing social organization. It does not claim the construction of a non-empty concrete model of an organizational ACS, verification of all conditions for an arbitrary real organization, or any necessity that ACS structure arises from organizing. It also does not treat a formal rule for non-reducibility that does not depend on a particular implementation, a universal procedure for judging the identity of an organization, or general social dynamics including the generation and withdrawal of members. That an organization qualifies as an organizational ACS does not mean the achievement of purposes, the feasibility of all participating individuals, actual satisfaction of the three principles, or the holding of ethical responsibility. The conditions under which sustainable coordination among ACSs and other agents holds are to be examined as the formal connection with ECIT in Phase 3 (subsec:phase_outlook, prob:ecit_full).

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.